

CN PHASE 1

MAC 1 — PRIVATE DNS SERVER

FULL EVIDENCE DOCUMENTATION

Team	Packet Forge
Domain	cn-capstone.test
Mac 1	cn-dns — 10.7.9.245 — en0
Mac 2 / Edge	10.7.5.53
DNS Service	dnsmasq on port 53
Evidence	Terminal + Wireshark + HTTPS verification

This document consolidates the supplied Mac 1 DNS configuration, DNS resolution tests, HTTPS verification and Wireshark packet captures into one compact evidence report.

1. DNS SERVER CONFIGURATION

Mac 1 (cn-dns) is the private DNS server. dnsmasq listens on localhost and the LAN address and binds to en0.

```
port=53
listen-address=127.0.0.1,10.7.9.245
bind-interfaces
no-resolv
server=8.8.8.8
domain-needed
bogus-priv
cache-size=1000
local-ttl=30
conf-dir=/opt/homebrew/etc/dnsmasq.d,*.conf
interface=en0
```

Project records:

```
address=/app.cn-capstone.test/10.7.5.53
address=/api.cn-capstone.test/10.7.5.53
```

2. VERIFIED REQUEST PATH

1	DNS	app.cn-capstone.test	→ 10.7.5.53
2	TCP	Client → 10.7.5.53:8443	SYN → SYN-ACK → ACK
3	TLS	SNI=app.cn-capstone.test	Handshake + encrypted data
4	HTTPS	https://app.cn-capstone.test:8443	HTTP 200 OK
5	Backend	nginx upstream	Backend A / Mac 3 healthy

Core verification:

```
dig @127.0.0.1 +short app.cn-capstone.test → 10.7.5.53
dig @127.0.0.1 +short api.cn-capstone.test → 10.7.5.53
dig @10.7.9.245 +short app.cn-capstone.test → 10.7.5.53
```

3. DNS Service and Port 53

```
adityarana@cn-dns ~ % echo "==== MAC 1 - DNS SERVICE ===="

echo
echo "----- DNSMASQ SERVICE -----"
brew services list | grep dnsmasq

echo
echo "----- DNS PORT 53 -----"
[sudo lsof -nP -iUDP:53 -iTCP:53
==== MAC 1 - DNS SERVICE ====

----- DNSMASQ SERVICE -----
dnsmasq    none          root

----- DNS PORT 53 -----
[Password:
COMMAND PID    USER    FD    TYPE          DEVICE  SIZE/OFF  NODE  NAME
dnsmasq  868  nobody    4u    IPv4  0x9e0b5371cf46d8fb      0t0    UDP  10.7.9.245:53
dnsmasq  868  nobody    5u    IPv4  0x5dc4db2cd751f42b      0t0    TCP  10.7.9.245:53 (LISTEN)
dnsmasq  868  nobody    6u    IPv4  0xc5aec416178b1e9b      0t0    UDP  127.0.0.1:53
dnsmasq  868  nobody    7u    IPv4  0xbb1fdbbbc505db672      0t0    TCP  127.0.0.1:53 (LISTEN)
adityarana@cn-dns ~ %
```

Figure 1. dnsmasq is running as root and is listening on port 53 on both 10.7.9.245 and 127.0.0.1 over UDP/TCP. This confirms Mac 1 is actively providing DNS service.

4. dnsmasq Configuration

```
adityarana@cn-dns ~ % echo "==== MAC 1 - DNS CONFIGURATION ====="

echo
echo "----- MAIN DNSMASQ CONFIG -----"
cat /opt/homebrew/etc/dnsmasq.conf

echo
echo "----- PROJECT DNS RECORDS -----"
cat /opt/homebrew/etc/dnsmasq.d/project.conf
==== MAC 1 - DNS CONFIGURATION =====

----- MAIN DNSMASQ CONFIG -----
port=53
listen-address=127.0.0.1,10.7.9.245
bind-interfaces
no-resolv
server=8.8.8.8
domain-needed
bogus-priv
cache-size=1000
local-ttl=30
conf-dir=/opt/homebrew/etc/dnsmasq.d,*.conf

----- PROJECT DNS RECORDS -----
address=/app.cn-capstone.test/10.7.5.53
address=/api.cn-capstone.test/10.7.5.53
adityarana@cn-dns ~ %
```

Figure 2. The configuration binds dnsmasq to 127.0.0.1 and 10.7.9.245, uses en0, and loads the project records. The records map app.cn-capstone.test and api.cn-capstone.test to 10.7.5.53.

5. DNS Resolution

```
adityarana@cn-dns ~ % echo "==== MAC 1 - DNS RESOLUTION ====="

echo
echo "----- app.cn-capstone.test -----"
dig @127.0.0.1 +short app.cn-capstone.test

echo
echo "----- api.cn-capstone.test -----"
dig @127.0.0.1 +short api.cn-capstone.test

echo
echo "----- LAN DNS ADDRESS -----"
[dig @10.7.9.245 +short app.cn-capstone.test
==== MAC 1 - DNS RESOLUTION =====

----- app.cn-capstone.test -----
10.7.5.53

----- api.cn-capstone.test -----
10.7.5.53

----- LAN DNS ADDRESS -----
10.7.5.53
adityarana@cn-dns ~ %
```

Figure 3. The terminal evidence shows `app.cn-capstone.test` → 10.7.5.53 and `api.cn-capstone.test` → 10.7.5.53. The configured LAN DNS address is 10.7.9.245.

6. Final HTTPS Test

```
10.7.0.33
adityarana@cn-dns ~ % echo "==== MAC 1 – FINAL HTTPS TEST ====="

[curl -i https://app.cn-capstone.test:8443/api/status
==== MAC 1 – FINAL HTTPS TEST =====
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Thu, 01 Oct 2026 10:05:22 GMT
Content-Type: application/json
Content-Length: 56
Connection: keep-alive
X-Backend: A
Cache-Control: max-age=60

{"backend": "A", "status": "healthy", "server": "Mac 3"}%
adityarana@cn-dns ~ % █
```

Figure 4. The project hostname is used for the HTTPS request. The response is HTTP/1.1 200 OK from nginx, with X-Backend: A and a healthy Backend A response from Mac 3. Cache-Control is max-age=60.

7. Wireshark DNS Capture

The image shows a Wireshark capture of DNS traffic. The top pane displays a list of packets, with packet 15 selected. The middle pane shows the details of the selected packet, which is a User Datagram Protocol (UDP) packet. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
3	4.358222	10.7.9.245	10.7.9.245	DNS	68	Standard query 0x4b68 HTTPS my.newtonschool.co
4	4.358378	10.7.9.245	10.7.9.245	DNS	68	Standard query 0xa24b A my.newtonschool.co
5	4.377948	10.7.9.245	10.7.9.245	DNS	225	Standard query response 0x4b68 HTTPS my.newtonschool.co CNAME dualstack.prod-newton-web-1126586614.ap-south-1...
6	4.378013	10.7.9.245	10.7.9.245	DNS	195	Standard query response 0xa24b A my.newtonschool.co CNAME dualstack.prod-newton-web-1126586614.ap-south-1.elb...
9	8.346382	10.7.9.245	10.7.9.245	DNS	91	Standard query 0xd0e2 HTTPS c08j2v3juf1i.edison-edlet.newtonschool.co
10	8.346543	10.7.9.245	10.7.9.245	DNS	91	Standard query 0x0aee A c08j2v3juf1i.edison-edlet.newtonschool.co
11	8.359038	10.7.9.245	10.7.9.245	DNS	248	Standard query response 0xd0e2 HTTPS c08j2v3juf1i.edison-edlet.newtonschool.co CNAME dualstack.newton-learning...
12	8.361539	10.7.9.245	10.7.9.245	DNS	218	Standard query response 0x0aee A c08j2v3juf1i.edison-edlet.newtonschool.co CNAME dualstack.newton-learning-141...
15	8.399048	127.0.0.1	127.0.0.1	DNS	81	Standard query 0x028f A app.cn-capstone.test OPT
16	8.399167	127.0.0.1	127.0.0.1	DNS	97	Standard query response 0x028f A app.cn-capstone.test A 10.7.5.53 OPT

Frame 15: Packet, 81 bytes on wire (648 bits), 81 bytes captured (648 bits) on interface lo0, Null/Loopback

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

User Datagram Protocol, Src Port: 51500, Dst Port: 53

Source Port: 51500

Destination Port: 53

Length: 57

Checksum: 0xfe4c [unverified]

[Checksum Status: Unverified]

[Stream index: 7]

[Stream Packet Number: 1]

[Timestamps]

UDP payload (49 bytes)

Domain Name System (query)

0000 02 00 00 00 45 00 00 4d 92 94 00 00 40 11 00 00E..M....@...
0010 7f 00 00 01 7f 00 00 01 c9 2c 00 35 00 39 fe 4c5.9.L
0020 02 8f 01 20 00 01 00 00 00 00 00 01 03 61 70 70app
0030 0b 63 6e 2d 63 61 70 73 74 6f 6e 65 04 74 65 73 ..cn-caps tone tes
0040 74 00 00 01 00 01 00 00 29 10 00 00 00 00 00 00 t.....)
0050 00

User Datagram Protocol (udp), 8 bytes

Packets: 24 · Displayed: 10 (41.7%) · Dropped: 0 (0.0%)

Profile: Default

Figure 5. Wireshark shows the DNS query for app.cn-capstone.test and the corresponding response containing A 10.7.5.53. This provides packet-level DNS evidence.

8. TCP Handshake and TLS ClientHello

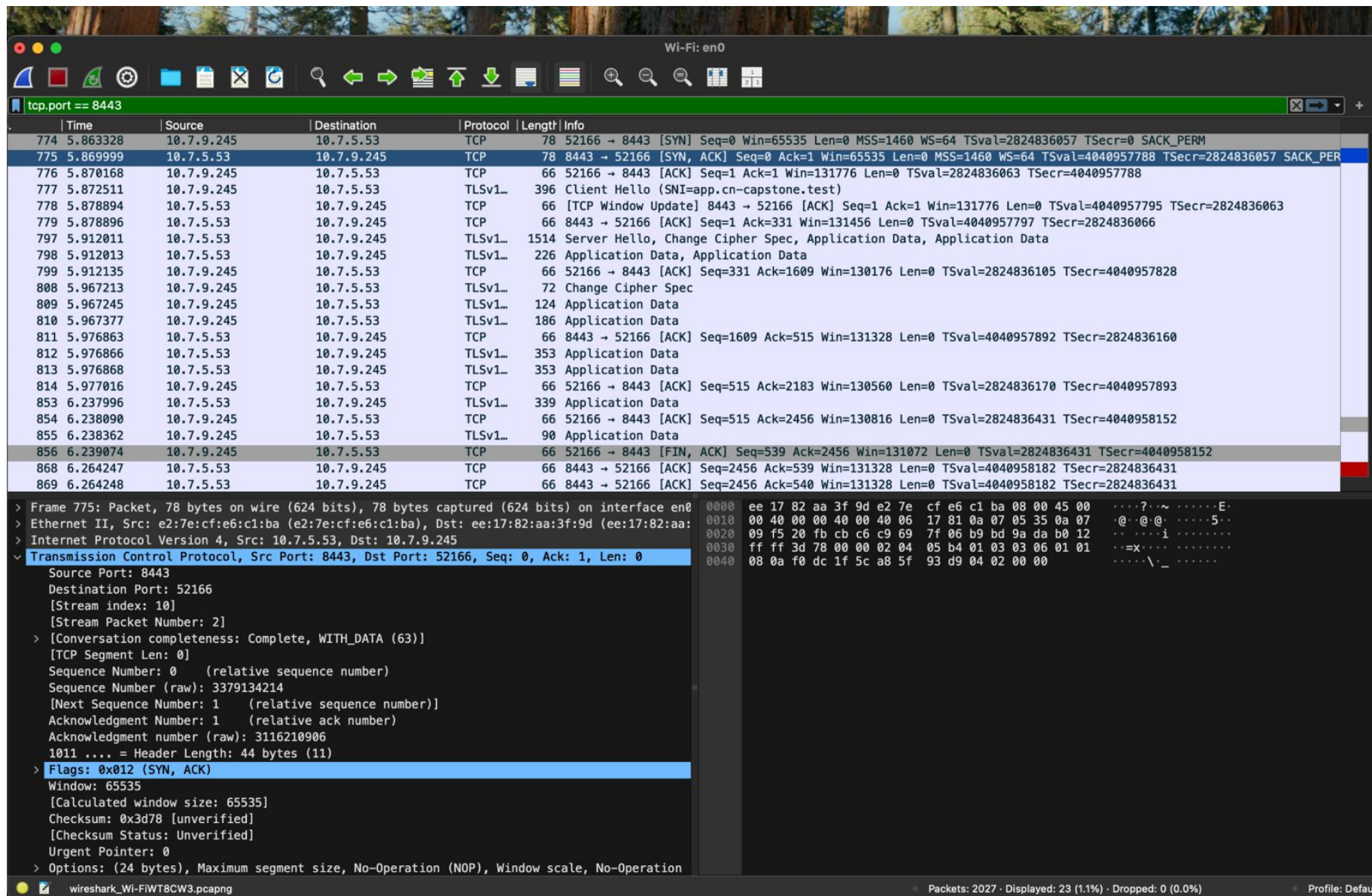


Figure 6. The capture shows SYN, SYN-ACK and ACK on TCP 8443, followed by a TLS Client Hello with SNI=app.cn-capstone.test. This proves the hostname reached the HTTPS edge and TLS negotiation started.

9. TLS Handshake and Encrypted Traffic

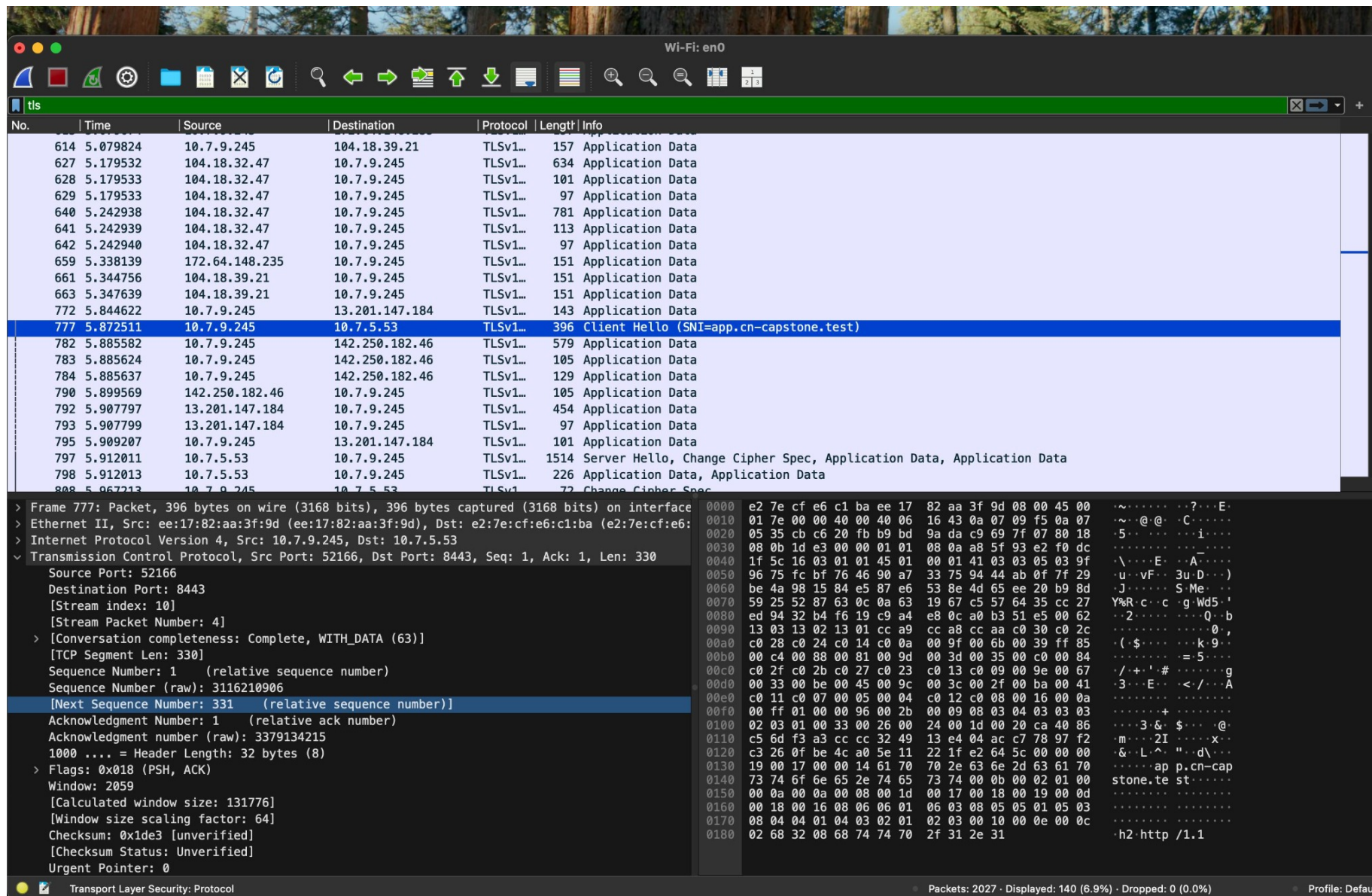


Figure 7. The capture shows Server Hello, Change Cipher Spec/Application Data and subsequent encrypted TLS traffic between 10.7.9.245 and 10.7.5.53.

10. TLS Application Data

Wi-Fi: en0

tls

No.	Time	Source	Destination	Protocol	Length	Info
663	5.347639	10.7.9.245	10.7.9.245	TLSv1...	151	Application Data
772	5.844622	10.7.9.245	13.201.147.184	TLSv1...	143	Application Data
777	5.872511	10.7.9.245	10.7.5.53	TLSv1...	396	Client Hello (SNI=app.cn-capstone.test)
782	5.885582	10.7.9.245	142.250.182.46	TLSv1...	579	Application Data
783	5.885624	10.7.9.245	142.250.182.46	TLSv1...	105	Application Data
784	5.885637	10.7.9.245	142.250.182.46	TLSv1...	129	Application Data
790	5.899569	142.250.182.46	10.7.9.245	TLSv1...	105	Application Data
792	5.907797	13.201.147.184	10.7.9.245	TLSv1...	454	Application Data
793	5.907799	13.201.147.184	10.7.9.245	TLSv1...	97	Application Data
795	5.909207	10.7.9.245	13.201.147.184	TLSv1...	101	Application Data
797	5.912011	10.7.5.53	10.7.9.245	TLSv1...	1514	Server Hello, Change Cipher Spec, Application Data, Application Data
798	5.912013	10.7.5.53	10.7.9.245	TLSv1...	226	Application Data, Application Data
808	5.967213	10.7.9.245	10.7.5.53	TLSv1...	72	Change Cipher Spec
809	5.967245	10.7.9.245	10.7.5.53	TLSv1...	124	Application Data
810	5.967377	10.7.9.245	10.7.5.53	TLSv1...	186	Application Data
812	5.976866	10.7.5.53	10.7.9.245	TLSv1...	353	Application Data
813	5.976868	10.7.5.53	10.7.9.245	TLSv1...	353	Application Data
846	6.189584	142.250.182.46	10.7.9.245	TLSv1...	1041	Application Data
847	6.189586	142.250.182.46	10.7.9.245	TLSv1...	250	Application Data
848	6.189587	142.250.182.46	10.7.9.245	TLSv1...	105	Application Data
850	6.190260	10.7.9.245	142.250.182.46	TLSv1...	101	Application Data
851	6.190338	10.7.9.245	142.250.182.46	TLSv1...	105	Application Data

> Frame 812: Packet, 353 bytes on wire (2824 bits), 353 bytes captured (2824 bits) on interface
> Ethernet II, Src: e2:7e:cf:e6:c1:ba (e2:7e:cf:e6:c1:ba), Dst: ee:17:82:aa:3f:9d (ee:17:82:aa:3f:9d)
> Internet Protocol Version 4, Src: 10.7.5.53, Dst: 10.7.9.245
✖ Transmission Control Protocol, Src Port: 8443, Dst Port: 52166, Seq: 1609, Ack: 515, Len: 287
Source Port: 8443
Destination Port: 52166
[Stream index: 10]
[Stream Packet Number: 14]
> [Conversation completeness: Complete, WITH_DATA (63)]
[TCP Segment Len: 287]
Sequence Number: 1609 (relative sequence number)
Sequence Number (raw): 3379135823
[Next Sequence Number: 1896 (relative sequence number)]
Acknowledgment Number: 515 (relative ack number)
Acknowledgment number (raw): 3116211420
1000 = Header Length: 32 bytes (8)
> Flags: 0x018 (PSH, ACK)
Window: 2052
[Calculated window size: 131328]
[Window size scaling factor: 64]
Checksum: 0x7745 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0

0000 ee 17 82 aa 3f 9d e2 7e cf e6 c1 ba 08 00 45 00?~.....E.
0010 01 53 00 00 40 00 06 16 e6 0a 07 05 35 0a 07 ...S...@...n...5..
0020 09 f5 20 fb cb c6 c9 69 85 4f b9 bd 9c dc 80 18 ...i...i...0...
0030 08 04 77 45 00 00 01 01 08 0a f0 dc 1f c4 a8 5f ...wE...2 Z.YX...u..
0040 94 40 17 03 03 01 1a 32 5a ac 59 58 f8 84 75 8d ...@...].] I..W...(
0050 cf df df ae cc 5d bc 5d 49 03 8f 57 94 da 28 8b ...+V...H...+Kg1
0060 a2 bf ac 2b 56 e5 04 48 f7 8c 2e 84 2b 4b 67 31 ...\$.|W...jQ
0070 1e 85 cd 24 eb 06 7c 57 ff fe eb 09 08 f2 6a 51 ...Z...Z...z
0080 16 02 d2 1d e8 81 ad 5a d4 8e d7 fa 28 c1 f6 7a f*qB..a.Z...TV..f
0090 66 2a 71 42 ad 61 83 5a 81 a2 9d 54 56 0c d2 66 ...p..Z..<..c9G'..
00a0 02 bc 81 70 a1 8b 5a a6 3c 9b c7 63 39 47 27 96 ...@...^Q...=0...^
00b0 c7 40 19 98 a6 5e 51 d7 09 a8 3d 4f 1e a8 d7 5e ...K...m &...D?
00c0 c2 06 4b 9a ae 07 bd 6d 26 8d d2 d7 c0 f9 44 3f ...\\..4Z..Cs ^...(
00d0 5c f8 07 34 5a e7 43 73 53 5e eb 28 97 1e 04 ca ...\\.:X..@...6..
00e0 1e 0d cb 5c b1 3a 58 81 40 be a3 e6 0c 36 8b bc ...~...~...J..2..
00f0 b4 1f 00 fc 7e d4 a2 e4 05 db 4a f4 32 f4 c3 de ...I4...*...U...
0100 49 34 13 e8 f9 9b e2 2a 94 d0 a4 55 7d aa ea 1c ...w..L..l...'
0110 90 0b 77 99 4c 7e 6c 03 8e c8 f4 60 ec 1d 08 27 ...c..p"L..3..Ib..
0120 b7 63 b7 70 22 4c bd 33 d7 49 62 85 07 89 d6 d3 ...=..rV...K...1..
0130 da 8c 3d c1 72 56 bf f5 e8 4b 12 d8 14 80 31 fd ...g..YA...M.Yae..
0140 b7 dc 67 9f 59 41 de 1d f1 b9 60 4d 59 61 65 9b ...\$[..H W8...c..
0150 db b8 0f cf 24 5b 12 48 57 38 06 89 fe 63 cd 9b ..
0160 8c

Transport Layer Security: Protocol

Packets: 2027 · Displayed: 140 (6.9%) · Dropped: 0 (0.0%)

Profile: Default

Figure 8. The selected packet is TLS Application Data from 10.7.5.53:8443 to the Mac 1 client. The TCP stream is established and the payload is encrypted.

11. Complete HTTPS Packet Flow

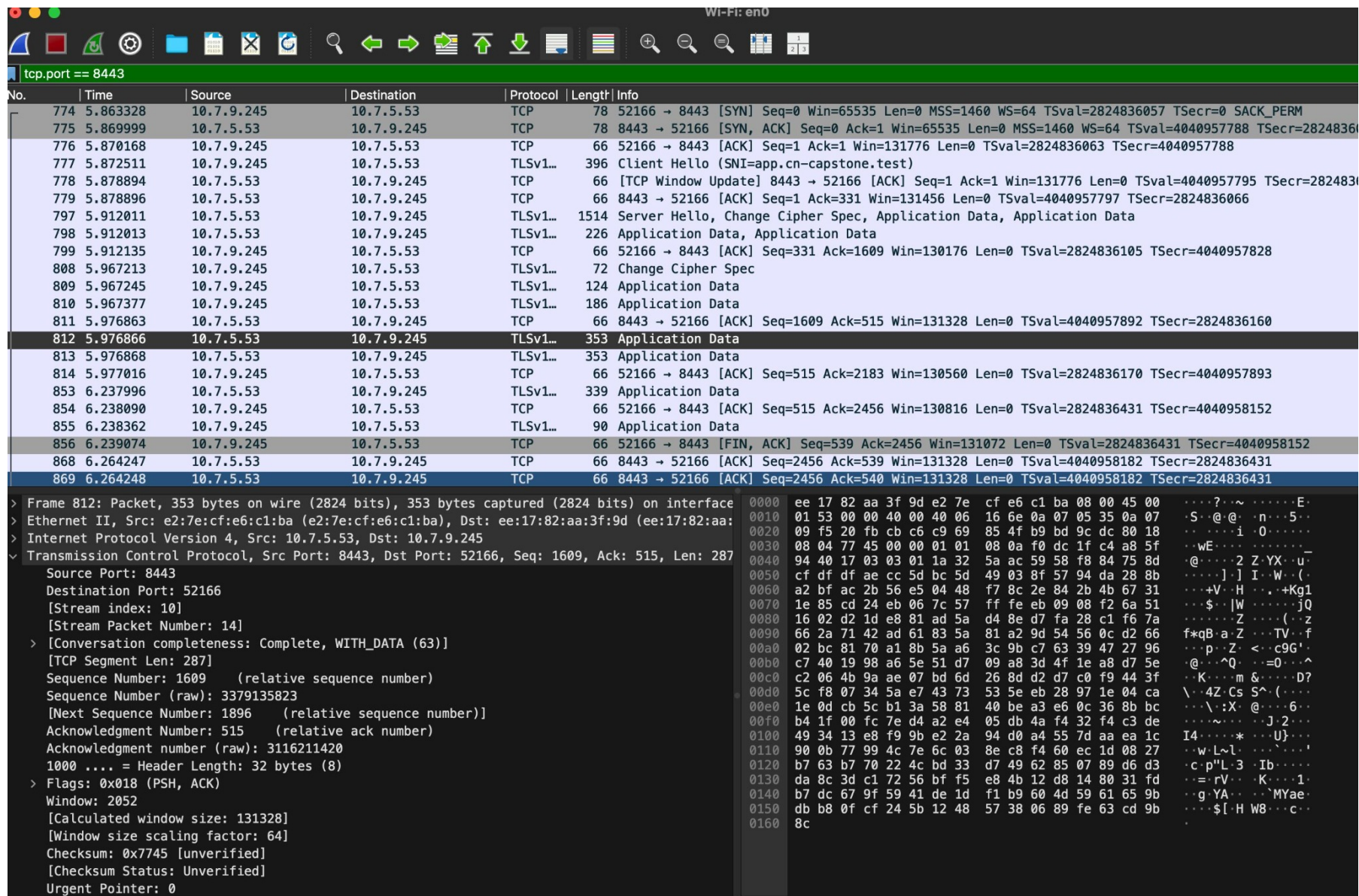


Figure 9. The capture provides the consolidated sequence: TCP setup, TLS Client Hello, Server Hello, encrypted Application Data, acknowledgements and connection termination.

12. FINAL EVIDENCE SUMMARY

Item	Evidence	Result
DNS server	dnsmasq / port 53	Verified on 10.7.9.245 and 127.0.0.1
Project DNS	app/api records	Both resolve to 10.7.5.53
DNS packets	Wireshark	Query + A response captured
TCP 8443	Wireshark	3-way handshake captured
TLS	Wireshark	ClientHello / ServerHello / encrypted data captured
HTTPS	curl	HTTP/1.1 200 OK
Backend	HTTP response	X-Backend: A; healthy Mac 3
Caching	HTTP response	Cache-Control: max-age=60

CONCLUSION

The supplied evidence demonstrates the complete documented path from the private DNS record on Mac 1 to the HTTPS application endpoint on Mac 2 and the healthy Backend A response on Mac 3. The DNS server configuration, port 53 listeners, DNS resolution, packet captures, TLS traffic and successful HTTP response are all included in this report.

Evidence scope: This report documents the supplied successful captures. It does not use those older captures to claim that any later managed-Mac firewall/TLS-reset issue was solved.